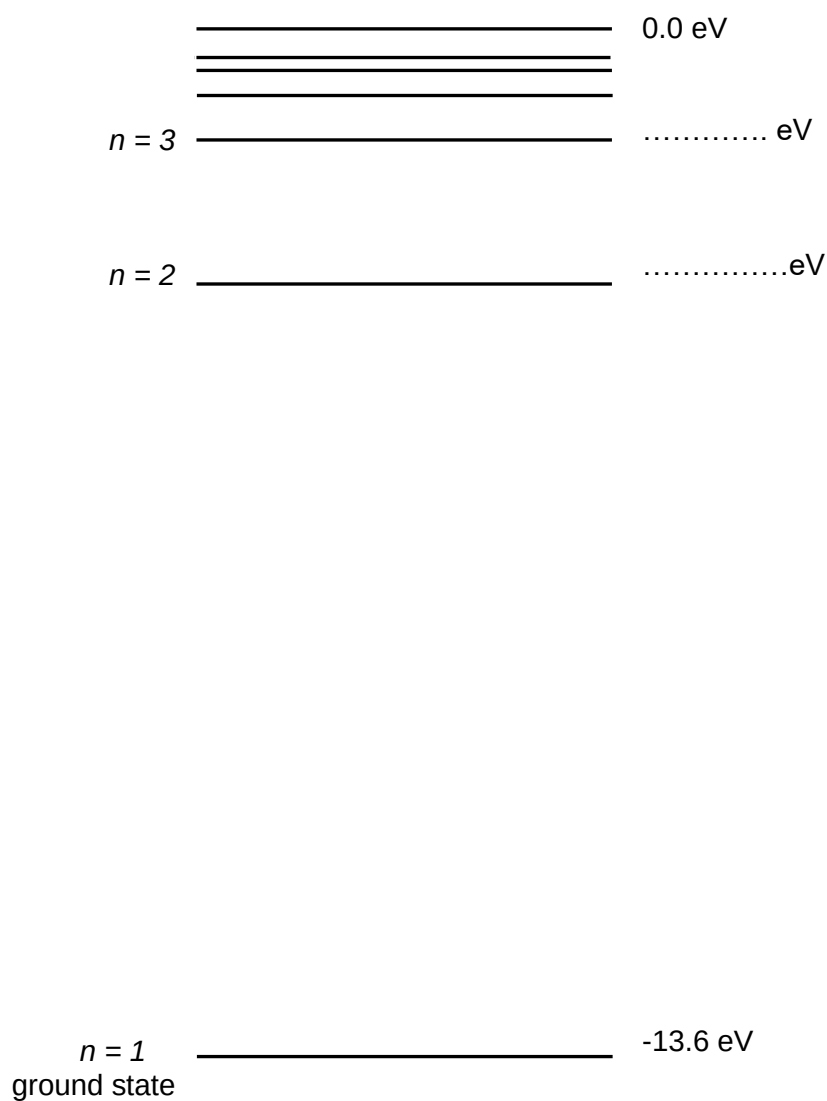


|   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                        |
|---|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 6 | (a) | State how a line emission spectrum may be explained on the basis of the existence of discrete electron energy levels in atoms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |
|   |     | <div><div>.....</div><div>.....</div><div>.....</div><div>.....</div><div>.....</div></div> <div>[3]</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                        |
|   |     | <div><div><b>Solution:</b></div><div>A line emission spectrum shows that only certain values of wavelengths (or frequencies) of photons are emitted when gas atoms de-excite.</div><div>Photons emitted have energies equal to the differences between the pairs of electron energy levels involved in the de-excitation.</div><div>The fact that only certain wavelengths are seen implies that only photons of certain energies are emitted, which in turn implies that electrons inside an atom can only possess certain discrete values of energies (called the electron energy levels).</div></div> | <div>1</div> <div>1</div> <div>1</div> |

|  |     |                                                                                                                                                                                                                                                                                                   |        |
|--|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|  |     |                                                                                                                                                                                                                                                                                                   |        |
|  | (b) | <p>The lowest six discrete energy levels for a hydrogen atom are shown in Fig. 6.1. The diagram is drawn to scale and the ground state is at -13.6 eV.</p>  <p style="text-align: center;"><b>Fig. 6.1</b></p> |        |
|  | (i) | On Fig. 6.1, label the values for energy at energy levels $n = 2$ and $n = 3$ .                                                                                                                                                                                                                   | [1]    |
|  |     | <p><b>Solution:</b></p> <p>Use ruler to measure</p> <p><math>n = 2</math> is - 3.4 eV<br/> <math>n = 3</math> is -1.5 eV</p>                                                                                                                                                                      | 1<br>1 |
|  |     |                                                                                                                                                                                                                                                                                                   |        |

|  |  |       |                                                                                                                                                                                                                                                                                                    |             |
|--|--|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|  |  | (ii)  | Calculate the wavelength of the light emitted when an electron falls from $n = 3$ energy level to the $n = 2$ energy level.                                                                                                                                                                        |             |
|  |  |       | wavelength = ..... m                                                                                                                                                                                                                                                                               | [3]         |
|  |  |       | <b>Solution:</b><br>Energy of photon = $(-1.5 - (-3.4))$ eV<br>$\frac{hc}{\lambda} = (-1.5 - (-3.4)) (1.6 \times 10^{-19})$<br>$\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\lambda} = (-1.5 - (-3.4)) (1.6 \times 10^{-19})$<br>$\lambda = 6.54 \times 10^{-7}$ m (For info: visible region) | 1<br>1<br>1 |
|  |  | (iii) | Electromagnetic radiation is emitted when an electron falls to the ground state from any of the other energy levels. State the region of the electromagnetic spectrum in which the radiation lies.                                                                                                 |             |
|  |  |       | electromagnetic radiation is .....                                                                                                                                                                                                                                                                 | [1]         |
|  |  |       | <b>Solution:</b><br>From (b)(ii), energy of photons emitted when electrons fall to $n = 2$ state lie in the visible region.<br>For greater difference in energy levels (eg. $n = 2$ to $n = 1$ , photons emitted when electrons fall to $n=1$ state lie in the ultraviolet region.                 | 1           |
|  |  | (iv)  | Describe one way by which an electron in gaseous hydrogen can be raised from a lower energy level to a higher energy level.                                                                                                                                                                        |             |
|  |  |       | .....<br>.....                                                                                                                                                                                                                                                                                     | [1]         |
|  |  |       | <b>Solution:</b><br>By thermal excitation by heating up the hydrogen gas. (electrons of high energy)<br>OR<br>By applying high voltage across the gas using a voltage discharge tube/ cathode ray tube<br>OR<br>Shine white light on the gas (use photons of all wavelengths)/ expose to sunlight  | 1<br>1<br>1 |

7

Read the passage below and answer the questions that follow.